THE MASTERMIND COACHING CLASSES

CBSE / RBSE · Nursery – Class XII · Road No. 5, VKIA, Jaipur · 9887258679

CLASS X — MATHEMATICS (Standard) · MID-TERM PRACTICE PAPER

Session 2026-27 · CBSE / Directorate of Education pattern · Sub code 041

Class: 10	Subject: Mathematics	Set 02	Time: 3 Hours	Max. Marks: 80

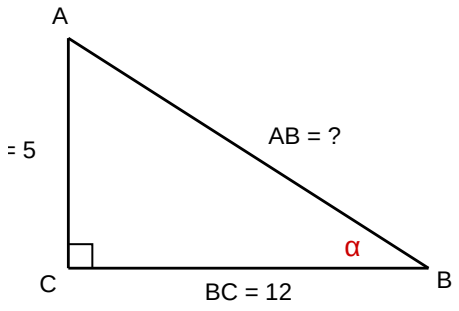
General Instructions:

1. This Question Paper has 5 Sections 'A', 'B', 'C', 'D' and 'E'.
2. Section A has 20 MCQs carrying 1 mark each (Q19–20 are Assertion–Reason questions).
3. Section B has 5 questions carrying 02 marks each.
4. Section C has 6 questions carrying 03 marks each.
5. Section D has 4 questions carrying 05 marks each.
6. Section E has 3 case based questions (04 marks each) with sub-parts of the values of 1, 1 and 2 marks respectively.
7. All questions are compulsory. However, an internal choice in 2 questions of 5 marks, 2 questions of 3 marks and 2 questions of 2 marks has been provided. An internal choice has been provided in the 2-mark questions of Section E. You have to attempt only one option out of the given options.
8. Draw neat figures wherever required. Take π = 22/7 wherever required, if not stated.
9. There is no provision of negative marking. Use of calculator is not permitted.
10. Please write down the serial number of the question before attempting it.

SECTION – A (Multiple Choice Questions · 20 × 1 = 20) — 20 questions · 20 marks

Q1–Q18 are multiple-choice questions. Q19–Q20: a statement of Assertion (A) is followed by a statement of Reason (R). Choose the correct option.

1. The value of cos α from the given figure is **1**



(a) $\frac{12}{13}$ (b) $\frac{5}{13}$ (c) $\frac{5}{12}$ (d) $\frac{13}{12}$

2. In ΔABC and ΔDEF, AB/DE = BC/EF and ∠B = ∠E. Then **1**

(a) ΔABC ~ ΔEDF (b) ΔABC ~ ΔDEF (c) ΔABC ~ ΔFED (d) ΔCBA ~ ΔDEF

3. The HCF and LCM of 15, 20 and 25 are respectively **1**

(a) 5, 150 (b) 1, 300 (c) 5, 300 (d) 10, 300

4. Two friends were born in a leap year. The probability that they have the same birthday is **1**

(a) $\frac{1}{365}$ (b) $\frac{1}{12}$ (c) $\frac{2}{366}$ (d) $\frac{1}{366}$

5. The distance of a point P from the *x*-axis is 5 units. Which of the following cannot be the coordinates of P? **1**

(a) (3, 5) (b) (5, 3) (c) (−3, −5) (d) (0, 5)

6. The value of $(4 + \sqrt{3})(4 - \sqrt{3}) - (\sqrt{5})^2$ is **1**

(a) 11 (b) 13 (c) 8 (d) 18

7. $\sqrt{8} + \sqrt{18} - \sqrt{50}$ equals **1**

(a) 0 (b) $2\sqrt{2}$ (c) $\sqrt{26}$ (d) $10\sqrt{2}$

8. A taxi charges ₹ 40 for the first kilometre and ₹ 12 for each additional kilometre. The fare for a 10 km journey is **1**

(a) ₹ 160 (b) ₹ 120 (c) ₹ 172 (d) ₹ 148

9. The pair of equations $kx + 2y = 7$ and $3x + y = 5$ has no solution when k equals **1**

(a) 2 (b) 3 (c) 6 (d) 7

10. The expression $(1 - \cos^2\theta)\,\text{cosec}^2\theta$ equals **1**

(a) 0 (b) 1 (c) $\sin^2\theta$ (d) $\text{cosec}^2\theta$

11. The mid-point of the line segment joining A$(2a, 4)$ and B$(-2, 2b)$ is $(1, 2a + 1)$. Then $a + b$ equals **1**

(a) 5 (b) 4 (c) 6 (d) 3

12. A 2-digit number less than 60 is selected at random. The probability that it is a multiple of 4 and 6 but not of 5 is **1**

(a) $\frac{1}{10}$ (b) $\frac{1}{12}$ (c) $\frac{4}{59}$ (d) $\frac{2}{25}$

13. The solution of the pair $3x + 2y = 13$ and $3x - 2y = 5$ is **1**

(a) x = 2, y = 3 (b) x = 4, y = 1 (c) x = 3, y = 2 (d) x = 1, y = 5

14. If x and y are different primes, the LCM of x^2y^3 and x^3y^2 is **1**

(a) x^3y^3 (b) x^2y^2 (c) x^5y^5 (d) x^3y^2

15. Two dice are thrown together. The probability that the difference of the numbers is 2 is **1**

(a) $\frac{1}{6}$ (b) $\frac{2}{9}$ (c) $\frac{5}{18}$ (d) $\frac{1}{9}$

16. If $\sin\theta = \frac{7}{25}$, then $\tan\theta$ equals **1**

(a) $\frac{24}{7}$ (b) $\frac{7}{18}$ (c) $\frac{25}{24}$ (d) $\frac{7}{24}$

17. The pair of linear equations $x + 2y = 4$ and $3x + 6y = 12$ represents two lines which are **1**

(a) parallel (b) coincident

(c) intersecting at one point (d) intersecting at two points

18. The common difference of the AP $\frac{1}{p}, \frac{1-p}{p}, \frac{1-2p}{p}, \ldots$ is **1**

(a) 1 (b) p (c) −p (d) −1

19. **Assertion (A):** The distance between the points (0, 0) and (−3, 4) is 5 units. **1**
Reason (R): The distance between (x_1, y_1) and (x_2, y_2) is $\sqrt{[(x_2 - x_1)^2 + (y_2 - y_1)^2]}$.

(a) Both A and R are true, and R is the correct explanation of A.

(b) Both A and R are true, but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

20. **Assertion (A):** For every angle θ, $(\sin\theta + \cos\theta)^2 = 1 + 2\sin\theta\cos\theta$. **1**
Reason (R): $\sin\theta + \cos\theta = 1$ for every angle θ.

(a) Both A and R are true, and R is the correct explanation of A.

(b) Both A and R are true, but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

SECTION – B (Very Short Answer Questions · 5 × 2 = 10) — 5 questions · 10 marks

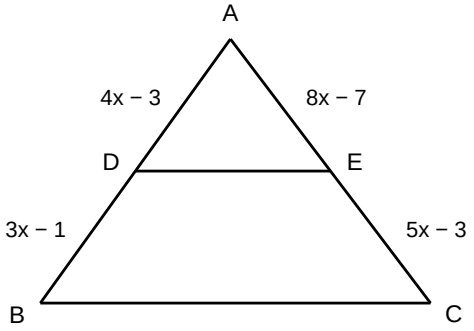
21. Find the HCF and LCM of 96 and 404 by the prime factorisation method. **2**

OR

Two traffic lights change after every 40 s and 60 s. If they change together at 8:00:00 a.m., when will they next change together?

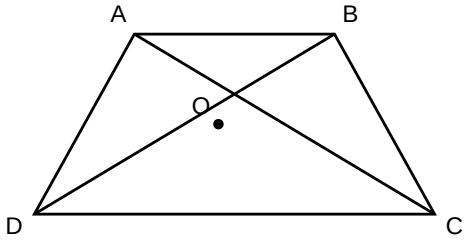
22. A girl 1.5 m tall casts a shadow 2 m long. At the same time a tree casts a shadow 12 m long. Find the height of the tree. **2**

23. In ΔABC, DE ∥ BC with AD = 4x − 3, DB = 3x − 1, AE = 8x − 7 and EC = 5x − 3. Find x. **2**



O R

In the trapezium ABCD (AB ∥ DC), the diagonals meet at O. Prove that OA/OC = OB/OD.



24. A ladder 10 m long just reaches a window 5√3 m above the ground. Find the angle the ladder makes with the ground. **2**

25. The LCM of two numbers is 2400 and their HCF is 20. If one number is 300, find the other. **2**

SECTION – C (Short Answer Questions · 6 × 3 = 18) — 6 questions · 18 marks

26. Prove that 5 + 3√2 is irrational, given that √2 is irrational. **3**

27. Solve for x and y: 7x − 3y = 31; 9x − 5y = 41. **3**

28. Find the sum of all two-digit numbers divisible by 7. **3**

O R

The sum of the first n terms of an AP is 3n^2 + 5n. Find its 20th term.

29. Find the point on the x-axis which is equidistant from (2, −5) and (−2, 9). **3**

O R

Find a relation between x and y such that the point P(x, y) is equidistant from A(5, 1) and B(−1, 5).

30. If 7 $\sin^2\theta$ + 3 $\cos^2\theta$ = 4, show that $\tan\theta = 1/\sqrt{3}$. **3**

31. Three coins are tossed together. Find the probability of getting (a) exactly two tails (b) at least two heads (c) no head. **3**

SECTION – D (Long Answer Questions · 4 × 5 = 20) — 4 questions · 20 marks

32. State and prove the Pythagoras theorem. **5**

33. A boat goes 30 km upstream and 44 km downstream in 10 hours; it goes 40 km upstream and 55 km downstream in 13 hours. Find the speed of the stream and the speed of the boat in still water. **5**

OR

Draw the graphs of $x + y = 5$ and $2x - y = 1$ on the same axes. Find their point of intersection and the points where they meet the y-axis.

34. From the top of a 60 m high building, the angles of depression of the top and the bottom of a tower are 30° and 60° respectively. Find the height of the tower. **5**

35. Prove that $\frac{\tan\theta + \sec\theta - 1}{\tan\theta - \sec\theta + 1} = \frac{1 + \sin\theta}{\cos\theta}$. **5**

OR

If $\tan(A + B) = \sqrt{3}$ and $\tan(A - B) = 1/\sqrt{3}$, $0° < A + B \le 90°$, $A > B$, find A and B. Hence evaluate $\sin(A + B) + \cos(A - B)$.

SECTION – E (Case Study Based Questions · 3 × 4 = 12) — 3 questions · 12 marks

36. At the sports-day lucky draw, 25 chits numbered 1 to 25 are placed in a box. A student draws one chit at random. Multiples of 4 win a badge, prime numbers win a pen, and a perfect square or a multiple of 10 wins a bonus ticket.

(i) Find the probability of winning a badge. 1

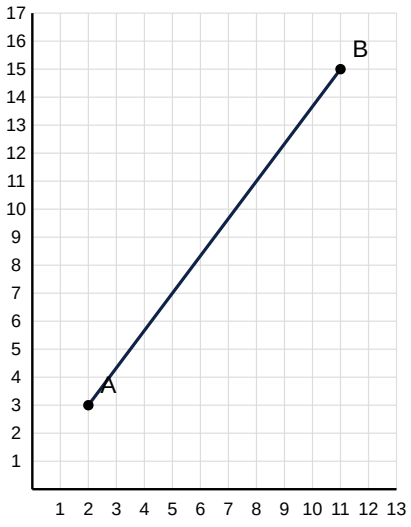
(ii) Find the probability that the chit drawn shows a prime number. 1

(iii) Find the probability of winning a bonus ticket. 2

OR

(iii) The first student draws chit 9 and keeps it. Find the probability that the next student draws a prime number.

37. A treasure-hunt map of a school ground is drawn on a coordinate grid (1 unit = 10 m). The flag is at A(2, 3) and the treasure box is at B(11, 15). Teams must walk straight from A to B.



(i) Find the distance AB in grid units. 1

(ii) Find the coordinates of the midpoint of AB. 1

(iii) A clue is hidden at point P on AB such that AP : PB = 1 : 2. Find the coordinates of P. 2

OR

(iii) Find the point on the x-axis which is equidistant from A and B.

38. Tarun is training for a 10 km run. On the first day he jogs 800 m and every day he increases the distance by 150 m. The distances covered on successive days form an AP.

(i) Write the first term and the common difference. 1

(ii) Write the expression for the distance he jogs on the nth day. 1

(iii) How many metres does he jog on the 8th day? 2

O R

(iii) What is the total distance he covers in the first 10 days?